A condition for this system to have a strong local minimum at $\delta \boldsymbol{p} = \mathbf{0}$, $\delta \boldsymbol{q} = 0$ is that the Hessian matrix $U''(\boldsymbol{q}^*)$ be positive definite. In case the eigenvalues of $U''(\boldsymbol{q}^*)$ are all distinct and positive, then the strong local minimum property will also follow for $\boldsymbol{q}^*$ in relation to the original potential.

1.6 Lattice Structures

Let us suppose we have a uniform pair potential φ and define the total potential energy of a system of N atoms by

$$U(x_1, x_2, \ldots, x_{\mathrm{N}}) = \sum_{i=1}^{\mathrm{N}-1} \sum_{j=i+1}^{\mathrm{N}} \varphi(|x_i - x_j|).$$

The computation of the energy requires $\mathrm{N}(\mathrm{N}-1)/2$ separate calculations, which could be very expensive if N is large. We can reduce this by assuming only *nearest neighbor* forces, which means the energy becomes

$$U(x_1, x_2, \ldots, x_{\mathrm{N}}) = \sum_{i=1}^{\mathrm{N}-1} \varphi(|x_{i+1} - x_i|).$$

While a Lennard-Jones interaction will maintain the particle ordering, a nearest neighbor system may otherwise require some care in the implementation, for example a mechanism for relabelling the particles when they cross. If the system is simulated without boundary conditions using short-ranged potentials like Lennard-Jones or Morse, then some atoms are likely to be ejected from the ends during simulation. In order to keep such a system bounded we might think to introduce walls at the ends of the chain, e.g. by adding confining potentials to U:

$$U(x_1, x_2, \ldots, x_{\mathrm{N}}) = \varphi_{\mathrm{c}}(|x_1|) + \varphi_{\mathrm{c}}(|L - x_{\mathrm{N}}|) + \sum_{i=1}^{\mathrm{N}-1} \varphi(|x_{i+1} - x_i|). \tag{1.7}$$

For example, one could assume $\varphi_{\mathrm{c}} = \varphi$ in which case we have effectively placed fixed particles at the ends of the interval, or one might use a different potential, e.g. a "soft wall" potential ($x^{-\alpha}$, e.g. $\alpha = 10$) or a spring (Cx^2). Alternatively one could restrict to a bounded domain by use of periodic boundary conditions, introducing the potential energy:

$$U(x_1, x_2, \ldots, x_{\mathrm{N}}) = \sum_{i=1}^{\mathrm{N}-1} \varphi(|x_{i+1} - x_i|) + \varphi(|L + x_1 - x_{\mathrm{N}}|). \tag{1.8}$$